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Pawpaw Database Roles and Privileges

Activity 4 Progress Documentation

1. Introduction

This document presents the progress of Activity 4: Database Users, Roles, Privileges, and Access Control for the Pawpaw Clinic project.

The activity focuses on creating database roles, assigning appropriate privileges, and testing access restrictions using PostgreSQL. The goal is to apply the principle of least privilege, where each database user receives only the permissions needed for their assigned responsibilities.

2. Project Overview

The project used for this activity is Pawpaw Clinic, which has a PostgreSQL database named group3_pawpaw.

The database contains the following tables:

- appointment
- appointment_service
- audit_log
- auth_session
- medical_record
- notification
- owner
- password_reset_code
- pet
- prescription
- service
- user_account
- vaccination
- veterinarian

The database was initialized using the project's migration and seed scripts and populated with sample records for testing.

3. Database Inspection

Before creating the roles, the database structure and existing records were inspected.

The database contains:

- 14 tables
- 12 sequences
- 1 schema named public

The following row counts were recorded:

Table	Number of Records
appointment	3
appointment_service	9
audit_log	40
auth_session	0
medical_record	1
notification	26
owner	3
password_reset_code	0
pet	2
prescription	0
service	10
user_account	7
vaccination	0
veterinarian	1

The inspection helped identify the existing database objects before assigning permissions.

4. Role Creation and Configuration

Two group roles were created based on the database access requirements of the Pawpaw Clinic application.

4.1 Group Roles

Group Role	Purpose
app_admin_grp	For database administration tasks
app_runtime_grp	For normal application operations

Both group roles were created with the NOLOGIN attribute. This means they cannot be used directly to connect to the database.

4.2 Login Roles

Login Role	Assigned Group Role
pawpaw_dba	app_admin_grp
pawpaw_app	app_runtime_grp

- The login roles were created with the LOGIN attribute and assigned to their corresponding group roles.
- Passwords were configured locally and were not included in the repository.

5. Privilege Assignment

Privileges were assigned according to each role's responsibility.

5.1 Database and Schema Privileges

Both group roles were granted CONNECT permission on group3_pawpaw. The app_runtime_grp and app_admin_grp roles were also granted USAGE permission on the public schema.

To restrict database access, the CONNECT privilege was revoked from PUBLIC on group3_pawpaw. Explicit CONNECT privileges were retained for the two intended group roles.

The CREATE privilege was also restricted on the public schema. Only the administrative group (app_admin_grp) was granted CREATE permission, while the runtime group (app_runtime_grp) received USAGE permission only.

5.2 Administrative Role

The app_admin_grp role was granted ALL PRIVILEGES on all existing tables and sequences in the public schema.

It was also granted USAGE and CREATE permissions on the public schema to allow administrative database operations.

The pawpaw_dba login role was assigned as a member of app_admin_grp to inherit its privileges.

5.3 Runtime Role

The app_runtime_grp role was granted:

- SELECT
- INSERT
- UPDATE
- DELETE

It was also granted USAGE and SELECT privileges on all sequences in the public schema.
The pawpaw_app login role was assigned as a member of app_runtime_grp to inherit its privileges.

The DELETE privilege was included in the runtime role's permissions, allowing the application account to perform deletion operations when needed.

5.4 Read-Only Role

No separate read-only group role was created for the Pawpaw Clinic database.
The database access configuration only includes two group roles:

- app_admin_grp – for administrative operations
- app_runtime_grp – for normal application operations

Read-only privileges were not assigned to a separate role because it was not included in the identified access requirements.

5.5 Default Privileges

- Default privileges were configured for future tables and sequences created by the postgres role in the public schema.
- The administrative group (app_admin_grp) receives ALL PRIVILEGES on future tables and sequences.
- The runtime group (app_runtime_grp) receives SELECT, INSERT, UPDATE, and DELETE privileges on future tables, and USAGE and SELECT privileges on future sequences.
- No default privileges were assigned to a separate read-only group because no read-only role was created.